Axel Sanchez

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Experience

AI Engineer, Scotiabank – Toronto, Ontario — Jun. 2026 – Present

- Founding engineer on **Scotia Intelligence**, the bank's central AI platform (**39K+ monthly active users**, **28M+ actions**); built its **Retrieval-Augmented Generation (RAG)** conversational search with inline citations over tools, FAQs, and training content
- Lead deployment of **4 agentic AI applications** on the bank's **LangGraph**-compatible agent harness across test, UAT, and production using **Kubernetes** (Argo CD, Helm), **Terraform**, and Microsoft Entra ID
- Developed an **LLM** Word-to-PowerPoint agent (**Python** agent, **TypeScript** UI) commissioned by the bank's Strategy Director; delivered reusable slide components and **4 executive slide templates**
- Shipped **4 multilingual knowledge agents** (English, French, Spanish) for procurement, accounting, and application-security guidance; added **CI/CD** testing, linting, and type checking
- Led an AI-assisted security scan of **21 applications** (**274 repositories**, **4.3M+ lines of code**), surfacing **642 vulnerabilities** to prioritize remediation
- Built a **Python** analytics and forecasting pipeline over **1.26M GCP Cloud Run** log records and proposed a **BigQuery** architecture; mentored an AI engineering intern on a **human-in-the-loop** RAG agent

Software Engineer Intern, Microsoft – Vancouver, British Columbia — May 2025 – Aug. 2025

- Designed and implemented custom **CUDA** AllReduce kernels for intra-node tensor parallelism on **NVIDIA H200** clusters; shipped them as a C++ **TensorRT-LLM** plugin with NVLink/P2P fallback
- Engineered zero-copy IPC user-buffer AllReduce with a **2× speedup over NCCL** on sub-2 MB payloads for tensor-parallel **LLM inference**; combined AllReduce with GEMM to overlap communication and computation
- Developed and benchmarked **C/CUDA BLAS** primitives (GEMM, AXPY, dot product, reduction) using tiled shared memory, loop unrolling, and IPC buffers across FP16/FP32
- Implemented **Triton** GPU kernels for broadcasting, activations, and AllReduce; built **Python** kernel-sanitization tooling to test correctness

Cloud Developer Intern, Aptum – Calgary, Alberta — May 2024 – Apr. 2025

- Managed production **Kubernetes** infrastructure with **Terraform** and Helm; optimized Jenkins CI/CD pipelines to **double deployment throughput**
- Refactored a legacy monolith into **gRPC microservices** with Spring Boot and MySQL, **reducing request latency by 66%** and enabling independent scaling
- Authored DevOps runbooks, Helm documentation, and onboarding guides adopted by **3+ intern cohorts**

Research

Undergraduate Researcher, University of Calgary – Calgary, Alberta — Sep. 2025 – Apr. 2026

- Developed **TheDataBay**, an Ethereum dataset marketplace with semantic **information retrieval** that ranks dataset metadata for natural-language queries using **LLM embeddings**; exposed REST APIs to React/TypeScript and React Native clients
- Built a privacy-preserving delivery pipeline that encrypted datasets before **IPFS** upload, stored integrity hashes on-chain, and released AES keys through **FastAPI** only after a **Solidity** smart contract verified buyer access

Projects

ChiselWare IP Factory Platform (Capstone) – *React, TypeScript, FastAPI, MySQL, AKS* — Rocksavage Technology Inc.

- Co-built a full-stack marketplace with a **6-engineer team** for verified submission, certification, and sale of Chisel-based semiconductor IP cores using **React/TypeScript**, **FastAPI**, and MySQL
- Deployed to **Azure Kubernetes Service** with Terraform, Container Registry, and Key Vault; automated builds and deployments through GitHub Actions CI/CD
- Integrated **Stripe** checkout with webhook-driven access grants, LinkedIn OIDC, and a Chisel-to-Verilog compilation worker; tested with pytest, Vitest, and **Playwright** E2E suites

Leadership

Founding Engineer, Agrivanna – Calgary, Alberta — Sep. 2024 – Apr. 2025

- Architected crop-health **REST microservices** in **Go** and Spring Boot; containerized with Docker and orchestrated on Kubernetes
- Productionized a **PyTorch** yield-forecasting model over **Kafka** streams; served predictions through FastAPI with GitHub Actions CI/CD

VP, Software Development, Student Energy UCalgary – Calgary, Alberta — Sep. 2023 – Apr. 2024

- Led a **6-engineer team** delivering a Next.js/TypeScript platform that **increased chapter engagement 70%**; owned AWS, Terraform, and GitHub Actions CI/CD
- Defined hiring rubrics, conducted **15+ interviews**, and authored succession documentation for the next executive team

Education

University of Calgary, Bachelor of Science in Software Engineering – Calgary, Alberta — Sep. 2021 – Apr. 2026

Skills

Languages: Python, C, C++, Java, JavaScript, TypeScript, Go, SQL

AI & ML: Generative AI, Agentic AI, LLMs, AI Agents, RAG, Embeddings, Vector Search, Prompt Engineering, LangGraph, LangChain, Gemini, PyTorch, TensorRT-LLM, CUDA, Triton

Frameworks: Spring Boot, FastAPI, React, React Native, Next.js

Cloud & Data: GCP, AWS, Azure, Docker, Kubernetes, Argo CD, Terraform, Helm, GitHub Actions, BigQuery, PostgreSQL, MySQL, Redis, Kafka

Systems & Testing: Data Structures & Algorithms, Linux/Unix, Distributed & Parallel Systems, Large-Scale System Design, Information Retrieval, Natural Language Processing, Networking, Security, Unit/E2E Testing

Certifications: Microsoft Certified Azure Fundamentals (2025), AWS Certified Cloud Practitioner (2024)